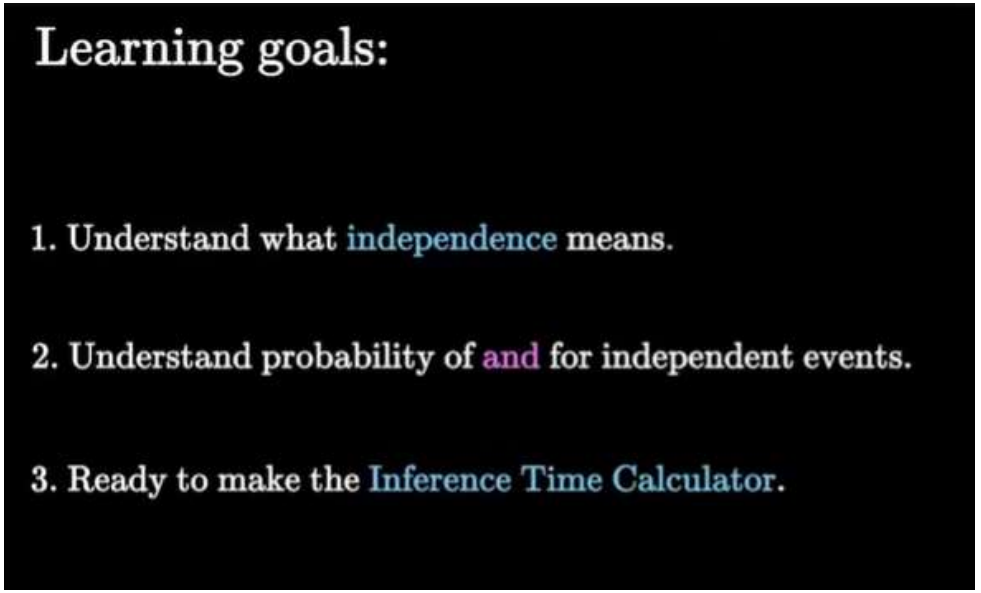
Independence
https://youtu.be/7WmJvrnJh-U?si=jm1MkPjGj25nii_m





The Stockholm study followed 739 women who were diagnosed with breast cancer within a year of a screening mammogram. A doctor read each mammogram, and an AI read the same images on its own.
Pick one of the 739 women at random. Let A be the event the AI misses her cancer and D the event the doctor misses it.
From the study: P(A)=0.181, P(D)=0.226, P(A and D)=0.114.
If A and D were independent, what would P(A and D) be?

Answer: 0.041 or 41%

If A and D were independent, we would multiply their probabilities:

$$P(A \text{ and } D) = P(A) \times P(D)$$

$$= 0.181 \times 0.226 = 0.040906$$

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From the study: P(A)=0.181, P(D)=0.226, P(A and D)=0.114. P(A and D)=0.114 is almost three times the 0.041 that independence predicts. The AI never saw the doctor's answer. It only saw the images.
Why might A and D still not be independent? Write a sentence or two.

Answer: The AI and doctor may both miss cancers that are difficult to see on a mammogram, even though they work independently. This means their mistakes are related because they are looking at the same images and some cancers are harder to detect than others.